Class 2

Quantum Field Theory II

Path Integrals, Renormalization, and the Standard Model

Duration: 3 months · 13 weeks · ~5–10 hrs/week
Part of the 2026–2027 Self-Study Plan · RHI → HEP Transition

Continues directly from QFT I. Covers path integrals, one-loop renormalization, non-Abelian gauge theory, and the electroweak sector. End state: able to read current HEP theory papers.

Course Overview

This class picks up where QFT I ends and is the single most important on the list for the RHI $\rightarrow$ HEP transition. Over 13 weeks you will learn path integrals, one-loop renormalization, the renormalization group, non-Abelian gauge theories, and enough of the electroweak sector and Higgs mechanism to read current ATLAS/CMS measurement papers.

The pedagogical line is Peskin & Schroeder Chs. 9–11, 15–17, and 20, with Schwartz used heavily for renormalization (his treatment is cleaner) and for the Standard Model chapters.

A heavy-ion physicist coming from QCD phenomenology will find much of this familiar at the level of results; the new content is the derivation of those results from first principles. Expect Weeks 3–6 (renormalization) to be the hardest.

Learning Goals

- Use the path integral formalism comfortably for scalars, fermions, and gauge fields.
- Compute one-loop corrections in QED: electron self-energy, vertex correction, photon vacuum polarization.
- Explain renormalization physically; derive the Callan-Symanzik equation; compute β functions.
- Quantize non-Abelian gauge theories including Fadeev-Popov ghosts and BRST.
- Derive asymptotic freedom in QCD; compute one-loop $\alpha_s(\mu)$.
- Explain the electroweak Lagrangian, spontaneous breaking, and the Higgs mechanism; identify every term and coupling.

Prerequisites & Background Check

This class assumes everything from QFT I. Do not start until you can derive QED Feynman rules from L_{QED} on a blank page.

- Completion of QFT I or equivalent Peskin Chs. 1–6.
- Comfort with trace technology, spinor sums, and tree-level amplitude computation.
- Gaussian integrals, including Grassmann variables (will be introduced in Week 2 but prior exposure helps).
- Group theory: SU(N) representations, structure constants, fundamentals to the Killing form level.

Textbooks & Core Resources

Primary Textbooks

- **Peskin & Schroeder**, Chs. 9–11, 15–17, 20. The spine.
- **Schwartz**, *Quantum Field Theory and the Standard Model*, Parts III–V. Strongly preferred over P&S for Weeks 4–7.

Supplementary / Reference

- **Weinberg**, Vol. II — for non-Abelian gauge theory and anomalies, consult as reference.
- **Srednicki**, Parts II–III — another angle on renormalization.
- **Thomson**, *Modern Particle Physics* — experimentalist-friendly SM; read in parallel with P&S Ch. 20.
- **Langacker**, *The Standard Model and Beyond* — modern SM reference.
- **Muta**, *Foundations of Quantum Chromodynamics* — dedicated QCD companion.

Online Lectures & Notes

- Tong's *Gauge Theory* and *Standard Model* notes (Cambridge).
- Iain Stewart MIT 8.323-8.324 lecture notes.
- Matthew Schwartz's Harvard QFT lecture videos (YouTube).
- TASI lectures on effective field theories (especially Manohar, Kaplan, and Luke sets).
- PDG reviews: “Quantum Chromodynamics,” “Electroweak Model and Constraints on New Physics,” “Status of Higgs-Boson Physics.”

Key Papers & Reviews

- Faddeev & Popov, *Feynman Diagrams for the Yang-Mills Field* (1967).
- ’t Hooft & Veltman, *Regularization and Renormalization of Gauge Fields* (1972).
- Gross & Wilczek; Politzer: asymptotic freedom papers (1973).
- Weinberg, *A Model of Leptons* (1967); Glashow (1961); Salam (1968).
- ATLAS and CMS Higgs discovery papers (2012) — read after Week 12.

13-Week Schedule

Hardest single block is Weeks 4–7 (renormalization). Budget extra hours those weeks and cut back on problem sets in the other parallel class if needed.

Week	Topic	Readings / Activities	Deliverable
1	Path integrals for scalars	P&S Ch. 9.1-9.2; Schwartz Ch. 14	Rederive KG propagator from PI
2	Fermionic path integrals; Grassmann	P&S Ch. 9.5; Schwartz Ch. 14	Gaussian integral with Grassmann
3	Path integrals for gauge fields	P&S Ch. 9.4	Set up QED PI w/ gauge fixing
4	One-loop in φ<super>4</super>, regularization	P&S Ch. 10.1-10.2; Schwartz Ch. 15-16	Compute φ<super>4</super> 1-loop
5	QED one-loop: vacuum polarization	P&S Ch. 10.3; Schwartz Ch. 16	Π(q<super>2</super>) at one loop
6	Electron self-energy & vertex	P&S Ch. 10.3; Schwartz Ch. 17-18	Lamb shift parametrics; compute g-2
7	Renormalization group; Callan-Symanzik	P&S Ch. 12.1-12.2; Schwartz Ch. 23	Derive β(e) at 1 loop
8	Non-Abelian gauge theory I	P&S Ch. 15.1-15.3	Derive covariant derivative & field st
9	Quantizing YM; Fadeev-Popov & BRST	P&S Ch. 16.1-16.3	QCD Feynman rules
10	QCD: asymptotic freedom	P&S Ch. 16.5, 16.7	Compute β(g_s) at 1-lo
11	Spontaneous symmetry breaking; Goldstone	P&S Ch. 11.1; Schwartz Ch. 28	Linear sigma model in Higgs phase
12	Higgs mechanism; electroweak unification	P&S Ch. 20.1-20.2; Thomson Ch. 15-17	Derive W, Z masses from L<sub>EW</su
13	Flavor, CKM, review	P&S Ch. 20.3; PDG SM review	Capstone writeup

Pacing assumes ~5–10 hrs/week on this course while running in parallel with one other class. If time is tighter, cut problem sets first; if time is looser, add problems and deeper paper reading.

Problem Sets & Exercises

Pick 1–2 problems per chapter; prioritize the ones that force actual loop calculations.

- P&S 9.1 (retarded propagator from PI).
- P&S 9.2 (Schwinger representation).
- P&S 10.2 (Alternatives to dimensional regularization).
- P&S 10.3 (Mass renormalization).
- P&S 11.2 (SUSY-like cancellations — foreshadows BSM).
- P&S 12.2 (β functions in Yukawa theory).
- P&S 15.2 (Arbitrary gauge parameter in QED).
- P&S 16.1 (Arnowitt-Fickler gauge).
- P&S 16.3 (1-loop gluon self-energy in QCD).
- P&S 20.2 (Higgs decays to fermion pairs).
- P&S 20.3 (Triple gauge couplings).

Capstone Project

Two-part writeup, ~10 pages. Part A is theoretical; Part B connects to a recent experimental result.

- Part A.1: Derive the one-loop β function of QCD from the gauge-fixed Lagrangian. Include ghost contributions.
- Part A.2: Solve the RG equation and plot $\alpha_s(\mu)$ from m_Z to 1 TeV. Compare to PDG values.
- Part B.1: Write down the full electroweak Lagrangian; identify every coupling in terms of g, g', v.
- Part B.2: Derive Higgs production cross section via gluon fusion at leading order and produce a numerical estimate at $\sqrt{s}$ = 13.6 TeV; compare to the latest ATLAS or CMS inclusive Higgs measurement.
- Part B.3: Briefly discuss one modern measurement (e.g., $H \to \gamma\gamma$ branching, or top Yukawa) — what loop physics enters, and how big are the corrections?

Milestones & Self-Assessment

At each checkpoint, you should be able to do the listed tasks without looking them up. If you can't, go back rather than pushing forward.

- **End of Week 3:** You can set up a path integral for a gauge theory with proper gauge fixing without referring to P&S.
- **End of Week 7:** You can explain renormalization physically, derive a one-loop counterterm, and explain what a β function is at the level of a colloquium audience.

- **End of Week 10:** You can quantize QCD from the Lagrangian, derive its Feynman rules including ghosts, and explain why it is asymptotically free.
- **End of Week 13:** You can read a CMS or ATLAS Higgs paper, follow the theoretical side, and explain to another heavy-ion physicist what λ, y_t, and v mean in the Standard Model.

End of class plan. Iterate after the first two weeks — what was over-ambitious gets cut; what felt thin gets deepened.